FAHIM AHMAD

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Education

- **Ahsanullah University of Science and Technology (AUST)** — Dhaka, Bangladesh
  B.Sc. in Industrial and Production Engineering — CGPA: 3.51 / 4.00 — *January 2019 – May 2023*

Deployed Systems

- **Robert Junior- Humanoid Robot** — *Geneva, Switzerland | 2026*
  Personal assistant humanoid robot
- **3-Robot AGV Fleet** — *Dhaka, Bangladesh | 2025 – Present*
  In live production at The Urmi Group RMG facility
- **EOD Tracked ROVs, Jontro Soinik v1 / v2** — *Mali 2024 | DR Congo 2025*
  Fielded in UN peacekeeping and humanitarian demining support

Research Interests

- Smart Manufacturing & Industry 4.0
- Additive Manufacturing
- Design for Manufacturing & Assembly
- Robotic Hardware Design & Manufacturing
- Humanoid & Field Robotics
- Data-driven Process Optimization

Research Experience

- **Undergrad Thesis:** Investigation of Ceramic Component Mechanical Properties Fabricated with FDM Process
  Supervisor: Md. Ashiquzzaman Nipu, Assistant Professor, Dept. of Industrial and Production Engineering, AUST.
  Fabricated zirconium silicate and PLA composite specimens ($ZrSiO_4$ 60%, PLA 40%) through FDM and evaluated their toughness and flexural strength across systematic variations in layer height, nozzle diameter, and infill density. Applied KNN, linear regression, and random forest models to predict process and property relationships and quantify prediction error. The study linked additive manufacturing process parameters to mechanical performance and was published in the IMEOM 2022 conference proceedings.

- **Industrial R&D:** Jontro Soinik 1.0, Man-transportable EOD ROV with 6-DOF Robotic Arm
  Supervisor: Md. Rashed Mujib Noman, CEO, Cybernetics Hi-Tech Solutions (Pvt.) Ltd.
  Conducted kinematic and dynamic analysis of a 6-DOF manipulator to derive workspace envelopes and torque requirements, and validated chassis and arm reliability under operational and shock loads through ANSYS structural simulation consistent with MIL-STD-810H. Designed and iterated 30+ mechanical components in SolidWorks with GD&T to a ±0.1 mm manufacturing tolerance while optimising mass distribution, packaging, and thermal management. FMEA applied across 10+ formal design reviews reduced late-stage design changes by 15%.

Publications

- **F.Ahmad**[*], M.N.Hasan[*], E.R.Bappy[*], M.M.Islam, **"Autonomous Material Handling in Bangladesh's Ready-made Garment Industry: An Exploratory Framework and Deployment Architecture for AGV/AMR Adoption."**
  Submitted to the 9th Bangladesh IEOM International Conference, Dhaka, Bangladesh, December 26–27, 2026.
  [*]Equal contribution.

- **F.Ahmad**, R.S.Karim, **"Machine Learning-Based Prediction of Mechanical Properties of Zirconium Silicate/PLA Ceramic Composites Fabricated via Fused Deposition Modeling."**
  Submitted to the 9th Bangladesh IEOM International Conference, Dhaka, Bangladesh, December 26–27, 2026.

- R.S.Karim, **F.Ahmad**, S.Z.Hasan, S.Shakur, Dr. M.A.Rahman, Dr. M.S.Bhuiyan, **"Investigation of Ceramic Component Mechanical Properties Fabricated with Fused Deposition Modeling Process."**
  Proceedings of the 5th International Conference on Industrial & Mechanical Engineering and Operations Management (IMEOM 2022), Dhaka, Bangladesh, December 26–27, 2022, pp. 568–569.

Professional Experience

- **RB Labs (Robot Bulls)** — Geneva, Switzerland
  Mechanical Design Engineer, Humanoid Robotics — *Aug 2026 – Present*
  - Design **mechanical systems and robot components for humanoid robot heads**, balancing industrial design aesthetics with structural and functional requirements
  - Develop **compact electronics packaging** that integrates sensors and control electronics efficiently within the head assembly
  - Conduct **R&D on materials and manufacturing processes** to optimise head assembly layouts
  - Collaborate with **engineering teams across Geneva and the USA** to accelerate and refine design iterations

- **Cybernetics Hi-Tech Solutions (Pvt.) Ltd.** Dhaka, Bangladesh
  Engineering Project Manager, Robotics & Automation *May 2023 – Present*
  Progression: R&D Engineer, Mechanical → Lead Mechanical Engineer → Engineering Project Manager
  Portfolio: EOD ROV | AGV | UAV | Smart Energy Meter | Industrial 3D Printer | Automation Systems

  - Directed **6 concurrent hardware product lines** across a **25 Crore BDT portfolio**, with full accountability for milestones, deliverables, and stakeholder reporting from R&D through field delivery
  - Managed **25 cross-functional engineers** across defense, energy, and industrial automation programmes
  - Reduced delivery lead time by **25%** through procurement restructuring and supply chain optimisation
  - Achieved **12% cost reduction** through BOM rationalization, vendor negotiation, and finance process control

  Project: EOD ROV "Jontro Soinik 1.0", Bangladesh Army & UN Peacekeeping Missions
  Role: R&D Engineer, then Mechanical Team Lead & PM | Team: 8 Engineers | Budget: 4 Crore BDT | Clients: DGDP / Bangladesh Army / UN Peacekeeping Mission

  - **Delivered 3 field-deployed EOD ROVs** to DGDP for the Bangladesh Army's Bomb Disposal Unit, accepted against military-grade criteria; units deployed on **UN Peacekeeping Missions in Mali and the Democratic Republic of Congo**
  - System formally **transferred to the Peruvian Armed Forces on 05 May 2024**, marking Bangladesh's first export of a domestically built defense robot to another nation's armed forces
  - Led the bid team securing government project approval: defined **120+ technical specifications and 30+ key deliverables** in full alignment with defense procurement authorities
  - Designed and validated actuation systems using **Harmonic and Planetary servo motors with CANBus protocols**; conducted kinematic analysis to meet torque and precision control requirements
  - Ensured compliance with **MIL-STD-810H, MIL-STD-461G, and MIL-STD-1472H** throughout design, validation, and risk assessment phases
  - Designed **30+ mechanical components and prototypes** in SolidWorks applying GD&T, achieving **±0.1mm manufacturing tolerance**
  - **FMEA-driven design validation** reduced development cycle time by **18%**; resolved **30+ cross-functional R&D blockers** spanning electronics, controls, and fabrication, reducing assembly error rate by **23%**
  - Facilitated **operational and maintenance training** for UN delegates and Peruvian Armed Forces personnel

  Project: Cyberfleet AGV System, RMG Factory Deployment
  Role: Project Manager

  - Designed **navigation logic and operational protocols** for AGV deployment in dynamic factory environments
  - Developed **emergency handling and obstacle-response strategies** aligned with autonomous operation
  - Achieved **13% reduction in material handling time** through sensor-optimised route redesign

  Project: UAV Multi-Payload Platform (Surveillance | Cargo 10 to 100 kg | Combat-Support)
  Role: Project Manager

  - Coordinated development of **autonomy-capable UAV platforms** across multiple payload classes
  - Managed **stakeholder engagement for 5+ clients** including Bangladesh Army and Border Guard Bangladesh
  - Reduced project costs by **10%** through BOM optimisation and resource allocation

Skills

Robotics & Systems: Robot system integration, humanoid robot hardware, 6-DOF kinematics, field testing, FMEA, structural analysis

Simulation & Modeling: ANSYS Workbench, SolidWorks Simulation, FEA

Design & Prototyping: SolidWorks (CSWP), Fusion 360, GD&T, DFM/DFA, electronics packaging, 3D Printing (FDM)

Standards: MIL-STD-810H, MIL-STD-461G, MIL-STD-1472H

Coding: Python, ROS (basic)

Project Management: MS Project, Jira, BOM Management, Risk Management, Stakeholder Management

Professional Certifications

- **SOLIDWORKS API Professional (CSWP-API)** November 2026
  Dassault Systèmes
- **SOLIDWORKS Simulation Professional (CSWP-Simulation)** November 2026
  Dassault Systèmes
- **SOLIDWORKS Sheet Metal Professional (CSWPA-SM)** November 2026
  Dassault Systèmes

- **SOLIDWORKS Weldments Professional (CSWPA-WD)** November 2026
  Dassault Systèmes
- **SOLIDWORKS Mold Making Professional (CSWPA-MT)** November 2026
  Dassault Systèmes
- **Lean Project Management Yellow Belt (LeanPM®-YB)** August 2025
  Lean Project Management Foundation
- **Lean Six Sigma White Belt (CSSC-LSSWB)** August 2025
  The Council for Six Sigma Certification
- **SOLIDWORKS CAD Design Professional (CSWP)** December 2024
  Dassault Systèmes

Achievements

- **1st Runner-Up** August 2022
  Innovate, IEOM Society AUST Student Chapter
- **Winner** June 2022
  Kambaii Bangabandhu Robotic Toolkit Design Competition, CRID Robotics Labs, USA
- **University Merit Scholarship, Level 3 Term 1** March 2022
  Department of Mechanical and Production Engineering, AUST *Spring 2021 Semester*
- **Top-5 Industrial Projects Award** December 2021
  Mujib-100 Industrial Exhibition, IC4IR
- **1st Runner-Up** March 2021
  TechnoCAD, Innoventure 21, AUST Innovation & Design Club

Extracurricular Activities

- **Vice President** Nov 2022 – May 2023
  AUST Innovation and Design Club
- **Affiliate** Nov 2022 – May 2023
  Computer Integrated Manufacturing (CIM) Lab, AUST
- **Organizer & Head of Sponsor Team** Mar 2022 – Jul 2022
  Mindsparks 22
- **Hardware & Design Team Member** Jan 2022 – Sep 2022
  AUST Rocketry
- **Mechanical Sub-Section Member & Design Lead** Sep 2021 – Sep 2022
  AUST Mars Rover Team, URC 2022, USA

References

Dr. Mohammad Sarwar Morshed
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